

Boolean Algebra

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Abstract:

In this paper, we explore Boolean algebra, its relationship with mathematical logic and rings. We prove every Boolean algebra is represented by a field of sets. We show every Boolean ring is a Boolean algebra. An immediate consequence restricts the cardinality of Boolean rings.

1. Propositional Calculus

Before delving into Boolean algebra, it may perhaps be more digestible to introduce some basic notion of propositional calculus.

Recall the following truth table (*T-schema*):

Table 1: Truth Table

p	q	$\neg p$	$p \vee q$	$p \rightarrow q$	$p \wedge q$	$p \leftrightarrow q$
1	1	0	1	1	1	1
1	0	0	1	0	0	0
0	1	1	1	1	0	0
0	0	1	0	1	0	1

Definition 1: Formal Systems

A **Formal System** is the following combination: **Formal Language** and **Deductive System**.

- **Alphabet:** A set of symbols commonly denoted Σ , containing both constants and variables.
- **Syntax:** A set of rules to dictate possible strings, called well-formed-formulas (wff).
- **Axioms:** A set of well-formed-formulas that are chosen to derive new theorems.
- **Inference Rules:** A set of rules that dictate the derivation of new theorems based on those established such as Substitution and Modus Ponus.

Both Alphabet and Syntax form a Formal Language. Both Axioms and Inference Rules form a Deductive System. It is noted that axioms are a subset of theorems. The derivation of new theorems goes through a structural induction. That is, you start with the smallest set, Axioms, and through Inference Rules derive new elements of Theorems.

Definition 2: Interpretations

An **interpretation** $\mathcal{I}$ is a recursive mapping from the set of wff dictated by the Formal Language to truth values.

Example: Let x be some proposition. $\mathcal{I}(x \wedge \neg x) = \min(\mathcal{I}(x), \mathcal{I}(\neg x)) = \min(\mathcal{I}(x), 1 - \mathcal{I}(x))$

Together both syntactic Definition 1 and semantic Definition 2 with the *T-schema* evaluates any wff with a truth value. There are some wff that under any interpretation will always have a constant truth value. Namely $p \vee \neg p$ which maps to 1 and $p \wedge \neg p$ which maps to 0.

Definition 3: Entailment and Equivalence

For any two wff P, Q , using the Deductive System, if Q is derivable from P then P **entails** Q . And if P is also derivable from Q then P and Q are **equivalent**.

This definition is far from the usual definition in logic where it includes interpretations. Here, the definition is purely syntactic.

2. Boolean Algebra

Definition 4: Boolean Algebra

Given any set B

B is a **Boolean algebra** iff a unary operation $\neg : B \rightarrow B$, called not

a binary operation $\vee : B \times B \rightarrow B$, called join

a binary operation $\wedge : B \times B \rightarrow B$ called meet

such that the following axioms are true:

$$\forall p, q, r \in B$$

$$(B1) \quad p \vee q = q \vee p \quad p \wedge q = q \wedge p$$

$$(B2) \quad (p \vee q) \vee r = p \vee (q \vee r) \quad (p \wedge q) \wedge r = p \wedge (q \wedge r)$$

$$(B3) \quad p \vee (p \wedge q) = p \quad p \wedge (p \vee q) = p$$

$$(B4) \quad p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r) \quad p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

$$(B5) \quad p \vee (q \wedge \neg q) = p \quad p \wedge (q \vee \neg q) = p$$

Think of B as a set of equivalence classes of wff and $p \in B$ is not some wff. In fact, a wff, say x , is an element of p . Then, inside $\neg p \in B$, we should find $\neg x$. This way, p is an equivalence class of the wff x and any equivalent statement belongs in p . Similarly, if $x \in p \in B$ and $y \in q \in B$, then the equivalence class of $x \vee y$ is $p \vee q$. As discussed earlier, let p be any element of the Boolean algebra, let $p \vee \neg p = 1$ and $p \wedge \neg p = 0$, we get two equivalence classes that exists in all Boolean algebra.

Example: $(\mathbb{Z}_2, \neg, +_2, \cdot_2)$

Here $\neg x = 1 +_2 x$ and it is a familiar Boolean ring which is a subset of Boolean algebra.

Note rings are in general not Boolean algebra. Consider $\mathbb{Z}$, if $\vee$ be $+$ and $\wedge$ be $\cdot$ or swapped, we get $14 = 2 + (3 \cdot 4) = (2 + 3) \cdot (2 + 4) = 30$ by B4. Then, $30 - 14 = 16 = 0$ but $0 \in \mathbb{Z}$ is unique.

Example: Let X be any set. $(\mathcal{P}(X), \neg, \cup, \cap)$ is a Boolean algebra.

Here $\neg A = X \setminus A$. In fact, $(\mathbb{Z}_2, \neg, +_2, \cdot_2)$ is a specific example where X is a singleton set. Indeed, this is the most general type of Boolean algebra as any other Boolean algebra is *represented* by the constraints on X or as a subalgebra of $\mathcal{P}(X)$.

Similar to rings, Boolean subalgebras are Boolean algebras that are a subset and retains 0 and 1 from the parent Boolean algebra.

Definition 5: Boolean Homomorphisms

For any A, B Boolean algebras. For any $\varphi : B \rightarrow A$

φ is a **A-valued homomorphism** iff

$$(1) \quad \varphi(\neg p) = \neg \varphi(p)$$

$$(2) \quad \varphi(p \vee q) = \varphi(p) \vee \varphi(q)$$

$$(3) \quad \varphi(p \wedge q) = \varphi(p) \wedge \varphi(q)$$

Example: Let B be any Boolean algebra. Consider the 2-valued homomorphism $\mathcal{I} : B \rightarrow \mathbb{Z}_2$. This is an interpretation homomorphism on B . It is direct to verify it always maps 0 to 0 and 1 to 1.

Definition 6: Boolean Ideals

For any B , Boolean algebra. For any $I \subseteq B$ I is an **ideal** of B iff

- (1) $0 \in I$
- (2) $p, q \in I$ then $p \vee q \in I$
- (3) $p \in I, q \in B$ then $p \wedge q \in I$

Indeed, this looks very similar to the ideals as defined in rings.

Example: (0) is the trivial ideal of any Boolean algebra.

Example: Let φ be the 2-valued homomorphism. Its kernel is an ideal.

Lemma 1: Ideal are kernels

For any ideal $I \subseteq B$, there exists a homomorphism φ such that $\ker(\varphi) = I$.

Proof: The proof is identical to that of rings.

Definition 7: Maximal Ideals

For any $M \subseteq J \subseteq B$ ideals M is a **maximal ideal** of B iff

$$M \subset J \text{ then } J = B \\ \text{or } J \subset B \text{ then } M = J$$

Theorem 1: Maximal Ideal Theorem

Every proper ideal I of a Boolean algebra B is a subset of a maximal ideal M .

Proof: Suppose $|B| \leq \aleph_0$ then let $B = \{b_0, b_1, \dots\}$. Let $I_0 \subseteq I_1 \subseteq \dots$ be a sequence of ideals. For the base case, suppose $I = I_0 = M$ then we're done. In the induction step, Suppose I_n is defined. If $\neg b_n \in I_n$ then let $I_{n+1} = I_n$. Otherwise, let I_{n+1} be generated by $I_n \cup \{b_n\}$. In either case, we get

$$I \subseteq M = \bigcup_{n \in \mathbb{N}} I_n$$

which is maximal because, if any $b_n \notin M$ is added to M , then both $b_n, \neg b_n \in M$. Consequently, $b_n \vee \neg b_n = 1 \in M$ but $(1) = B$. If $|B| > \aleph_0$, the proof requires the Axiom of Choice.¹

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Theorem 2: Existence Theorem

For any B Boolean algebra. For any $0 \neq p \in B$. There exists a 2-valued homomorphism φ such that $\varphi(p) = 1$.

Proof: By Lemma 1, this is equivalent to showing there exists an ideal, therefore a maximal one M by Theorem 1 such that $p \notin M$. Let I be constructed using the condition $q \in I$ iff $(\neg p) \wedge q = q$. A short verification shows the first two conditions of an ideal are satisfied. For the third condition, suppose $q \in I, r \in B$, if $r \wedge \neg p = \neg p$ then for any $q \in I, r \wedge q = r \wedge (\neg p \wedge q) = (r \wedge \neg p) \wedge q = \neg p \wedge q = q \in I$. Otherwise, $r \wedge \neg p = r$ then $r \in I$ by construction. Then, $p \notin M$, else $1 \in M$.

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¹[1] Proof with Axiom of Choice is presented here.

Definition 8: Field of Sets

Let X be any set. Let $\mathcal{F}$ be a family of subsets of X . $(X, \mathcal{F})$ is a **field of sets** iff

- (1) $F \in \mathcal{F}$ then $X \setminus F \in \mathcal{F}$
- (2) $\emptyset \in \mathcal{F}$
- $\forall n \in \mathbb{N}$ (3) $\bigcup_{i < n} F_i \in \mathcal{F}$
- (4) $\bigcap_{i < n} F_i \in \mathcal{F}$

Note here the field of sets is not the familiar algebraic field and in general is not.

Example: $(X, \mathcal{P}(X))$ is a field of sets. Which is a Boolean algebra.

Example: Let $\mathcal{F}$ be all finite or cofinite subsets of X . Then, $(X, \mathcal{F})$ is a field of sets. In fact, it is also a Boolean algebra. This leads us to the next theorem.

Theorem 3: Stone's Representation Theorem

Any Boolean algebra is isomorphic to a field of sets.

Proof: Let B be any Boolean algebra. Let $X = \{x \mid x : B \rightarrow \mathbb{Z}_2 \text{ and } x \text{ is a homomorphism}\}$.
Let $f : B \rightarrow \mathcal{P}(X)$ such that

$$\forall p \in B \quad f(p) = \{x \in X \mid x(p) = 1\}$$

To verify it is a homomorphism:

$$\begin{aligned} \forall p \in B \quad f(\neg p) &= \{x \in X \mid x(\neg p) = 1\} \\ &= \{x \in X \mid \neg x(p) = 1\} \\ &= \{x \in X \mid x(p) \neq 1\} \\ &= X \setminus \{x \in X \mid x(p) = 1\} \\ &= X \setminus f(p) \end{aligned}$$

$$\begin{aligned} \forall p, q \in B \quad f(p \vee q) &= \{x \in X \mid x(p \vee q) = 1\} \\ &= \{x \in X \mid x(p) \vee x(q) = 1\} \\ &= \{x \in X \mid x(p) = 1 \text{ or } x(q) = 1\} \\ &= \{x \in X \mid x(p) = 1\} \cup \{x \in X \mid x(q) = 1\} \\ &= f(p) \cup f(q) \end{aligned}$$

Similar for $f(p \wedge q)$. To show injectivity, check $\ker(f) = \{p \in B \mid f(p) = \emptyset\}$. By Theorem 2, all non-zero elements of B must have a homomorphism that maps it to 1, then $\ker(f) = \{0\}$. To show surjectivity, let $S \subseteq X$. For every $x \in S$, pick all elements $p \in B$ such that $x(p) = 1$, meet all these elements and name it p_x . Then, $f(p_x) = \{x\}$ which leads to

$$\bigvee_{x \in S} p_x \in B \text{ and } f\left(\bigvee_{x \in S} p_x\right) = S$$

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3. Boolean Rings

Definition 9: Boolean Rings

For any ring R . R is a **Boolean ring** iff for every $x \in R$, $x^2 = x$.

Example: $(\mathbb{Z}_2, +_2, \cdot_2)$ is a Boolean ring.

Example: Let X be any set. $(\mathcal{P}(X), \Delta, \cap)$ where Δ is the symmetric difference, is a Boolean ring.

Lemma 2: Boolean Rings are Boolean Algebra

For any Boolean ring R , there exists a Boolean algebra B , R is definitionally equivalent to B .

Proof: Since for any $x \in R$ $x^2 = x$, then $-x = (-x)^2 = x^2 = x$. Then, $x + x = 0$. We wish to find some transformation that satisfy this property. Consider defining the plus operation of the Boolean ring with

$$+(x, y) = (x \wedge \neg y) \vee (\neg x \wedge y)$$

and meet operation of the Boolean algebra with the following

$$\vee (x, y) = x + y + xy$$

keeping

$$\wedge (x, y) = xy$$

A mechanical check shows transformations both ways satisfy Boolean ring and Boolean algebra conditions. Therefore they are definitionally equivalent.²

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Corollary 1: Cardinality of finite Boolean Rings

For any finite Boolean ring R , there is some n , an integer where $|R| = 2^n$

Proof: By Theorem 3, any finite Boolean algebra B is isomorphic to a finite power set. Then, the cardinality of Boolean algebra is always 2^n where n is an integer. By Lemma 2, any Boolean ring is a Boolean algebra and vice versa, then the cardinality of finite Boolean rings also has the same consequent.

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Bibliography

- [1] P. Halmos and S. Givant, "Logic as algebra," *The Dolciani Mathematical Expositions*, vol. 21, 1998, doi: 10.1090/dol/021.
- [2] P. Halmos and S. Givant, *Introduction to Boolean Algebras*. Springer New York, 2009. doi: 10.1007/978-0-387-68436-9.

²[2] As defined here.